Power BI: Maven Market Retail Performance Analysis

1. Executive Summary

Role: Business Intelligence Analyst
Tools: Microsoft Power BI, Power Query, DAX, Python (Seaborn)
Dataset: Maven Market (Multinational Grocery Retailer – US, Canada, Mexico)

The Maven Market Retail Performance Dashboard is a comprehensive end-to-end business intelligence solution designed to analyze the operational and commercial performance of a multinational grocery chain operating across North America.

By integrating transactional, customer, product, and store-level data, this solution delivers a 360° analytical view of revenue performance, profitability, customer demographics, product returns, and time-based growth trends.

The final report enables executives and regional managers to monitor KPIs, evaluate store performance, analyze customer behavior, and track revenue against strategic targets to support data-driven decision-making.

2. Business Problem & Objectives

Maven Market operates grocery stores of different sizes across the United States, Canada, and Mexico. The business required a centralized reporting solution to:

- Track KPIs across countries and regions
- Compare revenue against performance targets
- Monitor product returns and profitability
- Analyze customer growth and demographics
- Evaluate time-based revenue trends (YoY, MoM, YTD)

The raw data was distributed across multiple structured files including transactions, returns, customers, products, stores, and regional information. Without a unified data model, extracting insights was time-consuming and inefficient.

Key Objectives:

- Transform raw data into a professional star-schema model
- Build advanced DAX measures for commercial KPIs
- Develop interactive dashboards for executive and operational analysis
- Enable dynamic KPI selection and time intelligence tracking
- Deliver an executive-ready reporting experience with advanced UX design

3. The Process (Technical Implementation)

Step 1: Data Extraction & Transformation (ETL)

Import & Integration

- Connected to structured transactional datasets
- Integrated data from Transactions, Returns, Customers, Products, Stores, Regions, and Calendar lookup

Data Cleaning & Preparation (Power Query)

- Removed duplicates and handled missing values
- Standardized column names and data types
- Created calculated columns for demographic grouping (Age Groups, Education Level)
- Built a structured Calendar table to enable advanced time intelligence

Step 2: Data Modeling

Designed a robust **Star Schema** with 8 tables:

Fact Tables: Transactions & Returns

Dimension Tables: Customers, Products, Stores, Regions, Calendar, Measures Table

Key Modeling Features:

- One-to-Many relationships between dimensions and fact tables
- Proper filter propagation
- Optimized cardinality for performance
- Hidden foreign keys for clean report view

The model ensures scalability, performance optimization, and analytical clarity.

Step 3: Advanced DAX Calculations

Developed a comprehensive KPI framework including:

Core Measures

- Total Revenue
- Total Cost
- Total Profit
- Total Transactions
- Total Returns
- Quantity Sold
- Quantity Returned
- Unique Products

Efficiency & Performance Metrics

- Profit Margin %
- Return Rate %
- Revenue Target
- Revenue Target Gap
  Revenue Target % Improvement
- Revenue Target Gap % with Arrow
- Revenue Target Gap with Arrow

Time Intelligence

- YTD Revenue
- Last Month Revenue
- Last Month Profit
- Last Month Returns
- Last Month Transactions
- Last Year Revenue

- MoM Revenue %
- 60-Day Revenue
- % Weekend Transactions
- Weekend Transactions

Customer Metrics

- Total Customers (Purchasers)
- Last Month Customers (Purchasers)
- MoM Customers %

These measures allow dynamic comparisons, trend analysis, and executive-level KPI monitoring.

Step 4: Dashboard Design & Visualization

The final report consists of **5 interactive report pages** *(images section 6)*:

1. Executive Dashboard

High-level KPI overview including:

- Revenue vs Target
- Total Revenue, Profit, Transactions, Returns
- Revenue by Country
- Top 5 Stores by Revenue
- Top 5 Brands
- Top 10 Products

Designed specifically for senior leadership quick insights.

2. Regional Analysis

- Revenue breakdown by Country → Region → Store
- Drill-down capabilities
- Comparative performance across geographic hierarchies
- Store-level profitability insights

3. Product Detail

- Product-level revenue and profitability
- Brand impact analysis
- Return rate evaluation
- Dynamic KPI selection using Field Parameters

- Commercial metric comparison

4. Customer Detail

- Customer growth trends
- Revenue by Age Group
- Revenue by Gender
- Revenue by Education Level
- Customer segmentation
- Top customers by revenue contribution

Includes demographic-driven commercial analysis.

5. Time Intelligence Dashboard

- Year-over-Year Revenue Trends
- Month-over-Month Performance
- YTD Revenue Tracking
- Rolling 60-Day Revenue
- Seasonal insights

Provides a strategic view of performance evolution.

4. Advanced Dashboard Design & User Experience (UX)

This project emphasizes both analytics and user experience:

Custom Navigation

- Built a custom Page Navigation Bar using buttons and bookmarks
- Created seamless, app-like navigation
- Applied consistent color palette aligned with brand identity

Interactive Features

- Drill Up & Drill Down (Country → Region → Store, Year → Quarter → Month)
- Custom Tooltips for contextual micro-insights
- Managed Visual Interactions to preserve KPI integrity

- Dynamic Field Parameters to toggle between:
    - Revenue
    - Profit
    - Transactions
    - Returns

Advanced Visualizations (Python Integration)

To demonstrate extended analytical capability beyond default visuals:

- Heatmap (Revenue by Country vs Year/Month) using Seaborn

- Boxplot (Revenue distribution by Age Group)

This highlights flexibility in combining Power BI with Python-based analytics.

5. Key Insights Discovered

1. Overall Financial Performance (1998)

- **Revenue Performance:** Maven Market generated **$1.2M in total revenue**, exceeding its annual target by **+0.54%**, demonstrating stable commercial execution.

- **Profitability:** Total profit reached **$715,690**, resulting in a strong **59.68% profit margin**, indicating healthy cost control and pricing efficiency.

- **Seasonality: December** was the highest-performing month for both revenue and profit, with **November and December** showing peak performance. Only **January, September, November, and December** exceeded the revenue target, highlighting seasonal dependency.

2. Geographic & Store Performance

- **Country Contribution:**

    - **United States:** $0.61M (13 stores) – Highest total revenue contributor

    - **Mexico:** $0.48M (9 stores) – Strong performance with fewer stores

    - **Canada:** $0.11M (2 stores) – Smaller footprint, lower contribution

- **Top Performing Stores:**

    - Store 12 (Mexico) – Highest revenue and profit
    - Store 8 (Mexico)
    - Store 19 (Canada)
    - Store 13 (USA)

- **Operational Insight:** Certain **mid-sized stores outperform larger locations in profit margin efficiency**, suggesting that scale does not automatically translate to higher profitability. This presents an opportunity to replicate high-efficiency store models.

3. Product & Brand Performance

- **Top Revenue-Generating Brands:** Hermanos, Tell Tale, Ebony, Tri-State, and High Top
- **Top Revenue Product:** Great Pumpernickel Bread
- **Highest Profit Product:** Booker Strawberry Yogurt

Product performance indicates strong contribution from staple grocery categories, while profitability varies across SKUs—highlighting opportunities for margin optimization and pricing strategy refinement.

4. Customer Insights & Revenue Concentration

- **Top Customer:** Merridee Archuleta generated the highest number of orders and revenue contribution.
- **Revenue Concentration Risk:**
  The **65+ age segment contributes approximately 75% of total revenue across all three countries**.

This indicates:

- A structural dependency on senior customers
- High retention importance for this demographic
- Strong upsell and cross-sell potential due to high-value outliers

Strategic Considerations:

- Prioritize retention and loyalty initiatives for 65+ customers
- Expand targeted campaigns for the 55–64 segment to drive growth

- Reassess engagement strategies for 45–54 to increase contribution and reduce concentration risk

5. Weekly Performance Patterns

- **Top Revenue Days:**
    - Thursday (15.11%)
    - Monday (15.03%)
    - Wednesday (14.87%)

These days form a strong mid-week revenue cluster.

- **Weekend Performance:**
  Saturday and Sunday (~13.8%) perform consistently but below peak weekdays.
- **Lowest Performing Day:**
  Tuesday (13.48%), though variance across the week remains relatively balanced.

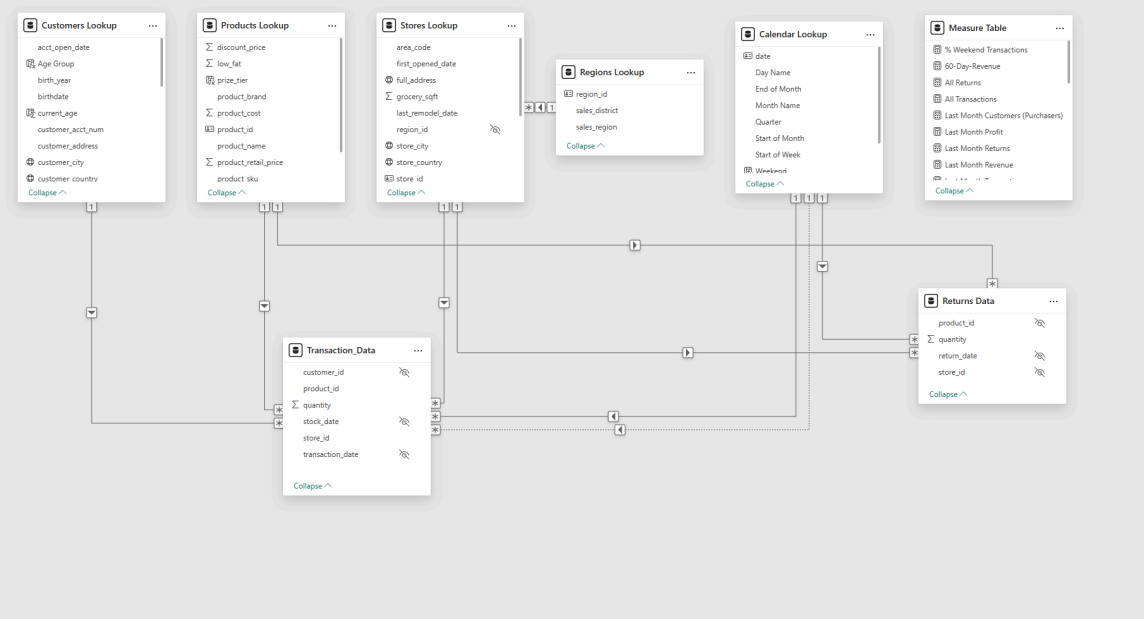
Operational Implication:
Staffing, promotions, and inventory optimization strategies should be aligned with mid-week demand peaks, while targeted promotions could be deployed to elevate lower-performing days.

Executive-Level Summary Insight

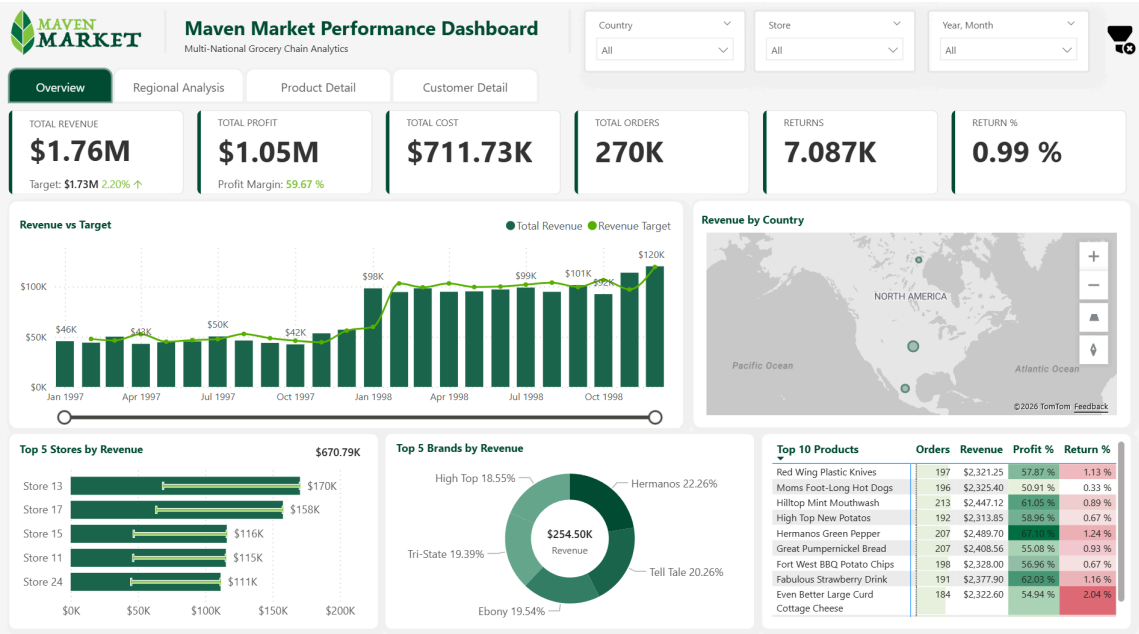
Maven Market demonstrates strong profitability and consistent target achievement, but performance is highly seasonal and significantly dependent on the 65+ customer segment. While the U.S. leads in revenue volume, Mexican stores exhibit strong individual store performance. Strategic focus should be placed on demographic diversification, seasonal planning, and replicating high-efficiency store models to sustain long-term growth.

6. Visuals

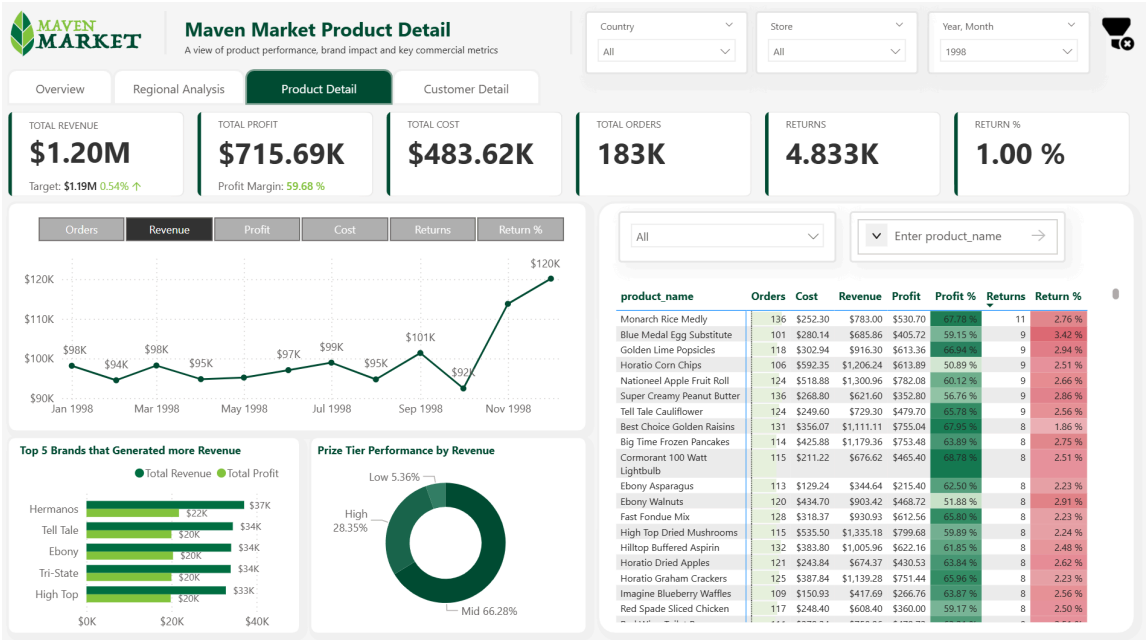
- Data Model (Star Schema)



- Executive Dashboard

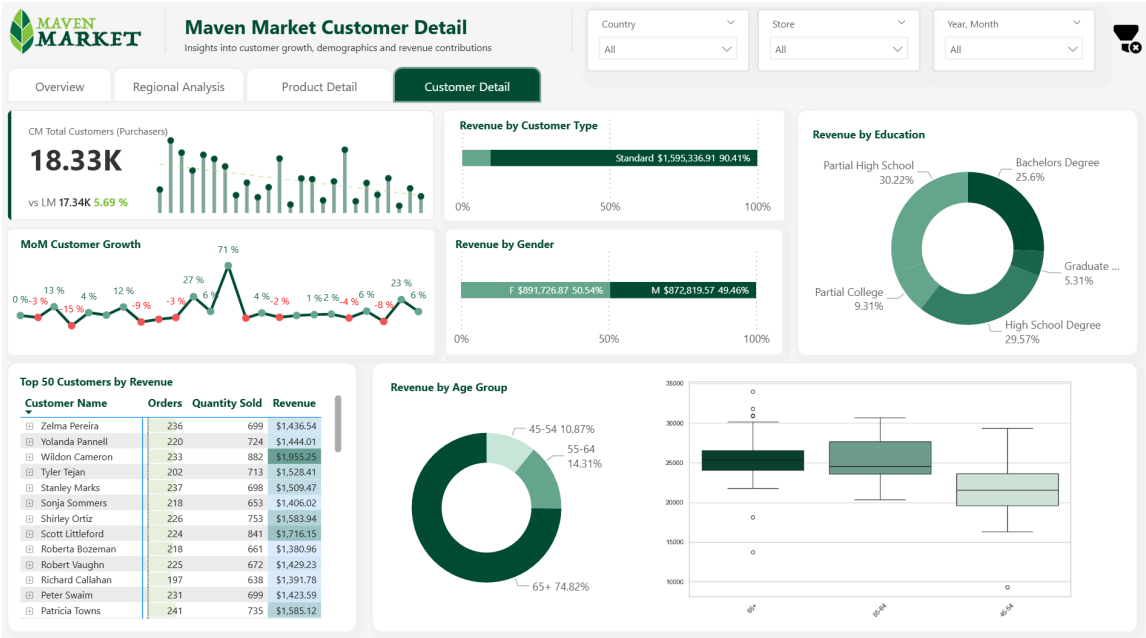


- Product Detail Page



product_name	Orders	Cost	Revenue	Profit	Profit %	Returns	Return %
Monarch Rice Medly	136	$252.30	$783.00	$530.70	67.78 %	11	2.76 %
Blue Medal Egg Substitute	101	$280.14	$685.86	$405.72	59.15 %	9	3.42 %
Golden Lime Popsicles	118	$302.94	$916.30	$613.36	66.94 %	9	2.94 %
Horatio Corn Chips	106	$592.35	$1,206.24	$613.89	50.89 %	9	2.51 %
Nationeel Apple Fruit Roll	124	$518.88	$1,300.96	$782.08	60.12 %	9	2.66 %
Super Creamy Peanut Butter	136	$268.80	$621.60	$352.80	56.76 %	9	2.86 %
Tell Tale Cauliflower	124	$249.60	$729.30	$479.70	65.78 %	9	2.56 %
Best Choice Golden Raisins	131	$356.07	$1,111.11	$755.04	67.95 %	8	1.86 %
Big Time Frozen Pancakes	114	$425.88	$1,179.36	$753.48	63.89 %	8	2.75 %
Cormorant 100 Watt Lightbulb	115	$211.22	$676.62	$465.40	68.78 %	8	2.51 %
Ebony Asparagus	113	$129.24	$344.64	$215.40	62.50 %	8	2.23 %
Ebony Walnuts	120	$434.70	$903.42	$468.72	51.88 %	8	2.91 %
Fast Fondue Mix	128	$318.37	$930.93	$612.56	65.80 %	8	2.23 %
High Top Dried Mushrooms	115	$535.50	$1,335.18	$799.68	59.89 %	8	2.24 %
Hilltop Buffered Aspirin	132	$383.80	$1,005.96	$622.16	61.85 %	8	2.48 %
Horatio Dried Apples	121	$243.84	$674.37	$430.53	63.84 %	8	2.62 %
Horatio Graham Crackers	125	$387.84	$1,139.28	$751.44	65.96 %	8	2.23 %
Imagine Blueberry Waffles	109	$150.93	$417.69	$266.76	63.87 %	8	2.56 %
Red Spade Sliced Chicken	117	$248.40	$608.40	$360.00	59.17 %	8	2.50 %

- Customer Detail Page



Customer Name	Orders	Quantity Sold	Revenue
Zelma Pereira	236	699	$1,436.54
Yolanda Pannell	220	724	$1,444.01
Wildon Cameron	233	882	$1,955.25
Tyler Tejan	202	713	$1,528.41
Stanley Marks	237	698	$1,509.47
Sonja Sommers	218	653	$1,406.02
Shirley Ortiz	226	753	$1,583.94
Scott Littleford	224	841	$1,716.15
Roberta Bozeman	218	661	$1,380.96
Robert Vaughn	225	672	$1,429.23
Richard Callahan	197	638	$1,391.78
Peter Swaim	231	699	$1,423.59
Patricia Towns	241	735	$1,585.12

- Time Intelligence Dashboard

Maven Market Time Intelligence

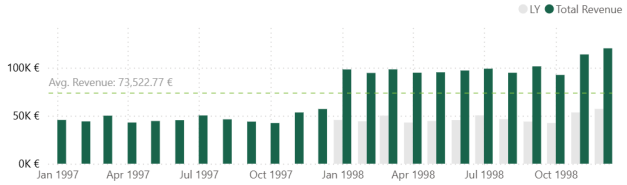
Time-based revenue insights across YoY trends, MoM performance and YTD tracking

Country: All | Store: All | Year, Month: All

Overview | Regional Analysis | Product Detail | Customer Detail

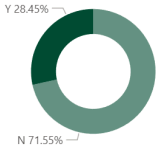
MTD REVENUE	MTD PROFIT	MOM GROWTH	MTD ORDERS	YTD REVENUE	60-Day-Revenue
$120.16K	$71.68K	5.60 %	18.33K	$1.20M	$229.89K
Target: **$119.48K** 0.57% ↑	Profit Margin: 59.66 %	vs Previous Month	Prev Month: **17.34K**		Rolling Period

YoY Comparison



MoM Revenue %

Revenue Weekend Analysis



Day Name	Revenue	% Total
Friday	$245,145.17	13.89%
Monday	$264,794.31	15.01%
Saturday	$250,260.66	14.18%
Sunday	$251,835.03	14.27%
Thursday	$266,162.12	15.08%
Tuesday	$234,129.83	13.27%
Wednesday	$252,219.32	14.29%

Year	Revenue	% of Parent	Rev. Target	Gap	YTD Revenue
⊟ 1997					
01/01/1997	$45,539.69	8%		$45,540 ↑	$45,539.69
01/02/1997	$44,058.79	8%	$47,816.67	-$3,758 ↓	$89,598.48
01/03/1997	$50,029.87	9%	$46,261.73	$3,768 ↑	$139,628.35
01/04/1997	$42,878.25	8%	$52,531.36	-$9,653 ↓	$182,506.60
01/05/1997	$44,456.29	8%	$45,022.16	-$566 ↓	$226,962.89
01/06/1997	$45,331.73	8%	$46,679.10	-$1,347 ↓	$272,294.62
01/07/1997	$50,246.88	9%	$47,598.32	$2,649 ↑	$322,541.50
01/08/1997	$46,199.04	8%	$52,759.22	-$6,560 ↓	$368,740.54
01/09/1997	$43,825.97	8%	$48,508.99	-$4,683 ↓	$412,566.51
01/10/1997	$42,342.27	7%	$46,017.27	-$3,675 ↓	$454,908.78
01/11/1997	$53,363.71	9%	$44,459.38	$8,904 ↑	$508,272.49
01/12/1997	$56,965.64	10%	$56,031.90	$934 ↑	$565,238.13
⊟ 1998					
01/01/1998	$98,155.28	8%	$59,813.92	$38,341 ↑	$98,155.28
01/02/1998	$94,498.00	8%	$103,063.04	-$8,565 ↓	$192,653.28
01/03/1998	$98,219.90	8%	$99,222.90	-$1,003 ↓	$290,873.18
01/04/1998	$94,776.14	8%	$103,130.90	-$8,355 ↓	$385,649.32
01/05/1998	$95,183.81	8%	$99,514.95	-$4,331 ↓	$480,833.13
01/06/1998	$97,050.04	8%	$99,943.00	-$2,893 ↓	$577,883.17
01/07/1998	$98,943.44	8%	$101,902.54	-$2,959 ↓	$676,826.61
01/08/1998	$94,718.51	8%	$103,890.61	-$9,172 ↓	$771,545.12
01/09/1998	$101,378.60	8%	$99,454.44	$1,924 ↑	$872,923.72